JAVA Talk 1

A: abelian variety of dim n

Hodge conj: (1941, formally presented in 1950)

$$\underbrace{H^{p,p}(A) \cap H^{2p}(A;\mathbb{Q})}_{Hdg^p(A) = B^p(A)} = \mathrm{Im}\left[\underset{\substack{\uparrow \\ \text{cycle class}}}{cl^p} : CH^p(A)_{\mathbb{Q}} \to H^{p,p}(A)\right]$$

$N^p(A)$

Know:

$$B^p(A) \supset \mathrm{Im}\, cl^p \supset D^p(A)$$

$N^p(A)$

"divisor class", $D^p(A) = \langle \mathrm{Pic}(A) \rangle_{alg}$

Thm [vG[redacted]94 Thm 4.2] $B^p(A) = D^p(A)$ in the following cases:

(1) (Mattuck 58) A is a general abelian variety;

(2) (Tate 65) $A = E_1 \times \cdots \times E_n$ $(\rightsquigarrow A^k)$

(3) (Tankeev 82) A simple, n is prime

(4) (Moonen-Zarhin 99) A simple, $n = 4$, X not of Weil-type.

Thm [vG94, Thm 4.11] (Weil 80)

For $n = 2m > 2$, A: general of Weil-type,

$$B^m(A) = \mathbb{Q}^3 \neq D^m(A) = \mathbb{Q}.$$

Still, Hodge conj is shown for some cases.

Thm [vG94, Thm 4.15] (Schoen 88)

For $n = 4$, A: general of Weil-type with $K = \mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, $\det H = 1$,

$$B^2(A) = \mathrm{Im}\, cl^2 = \mathbb{Q}^3.$$

Weil-type AV

Denote

$$CM_2 = \{\text{imaginary quadratic fields}\}$$
$$= \{\mathbb{Q}(\sqrt{-d}) \mid d \in \mathbb{Z}_{>0} \text{ squarefree}\}$$

$$\mathrm{End}^\circ(A) = \mathrm{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$$

($\varphi \in \mathrm{End}(A)$ fixes 0)

(I prefer to use $\mathrm{End}(A)_{\mathbb{Q}}$...)

<u>Def</u> (Weil-type) Let $n = 2m$ be even.

A is of Weil-type with CM field $K \in CM_2$, if

$$\exists\, \eta : K \hookrightarrow \mathrm{End}^\circ(A) \quad \text{s.t.}$$

$\forall x \in K$ (fix $K \subset \mathbb{C}$),

$$d \circ \eta(x) \sim \mathrm{diag}(\underbrace{x, \dots, x}_{m \text{ many}}, \underbrace{\bar{x}, \dots, \bar{x}}_{m \text{ many}})$$

i.e., the char poly of x on $T_0 A$ is

$$(T - x)^m (T - \bar{x})^m \in K[T].$$

Here,

$$\begin{tikzcd}
d \circ \eta : K \arrow[r, hook, "\eta"] & \mathrm{End}^\circ(A) \arrow[r, "d"] \arrow[dr, "d'"'] & \mathrm{End}_{\mathbb{C}}(T_0 A) \cong M^{n \times n}(\mathbb{C}) \\
& & \mathrm{End}_K(H^1(A;\mathbb{Q})) \cong M^{n \times n}(K) \\
& & \cap\!\!\shortmid \qquad\qquad \cap\!\!\shortmid \\
& & \mathrm{End}(H^1(A;\mathbb{Q})) \cong M^{2n \times 2n}(\mathbb{Q})
\end{tikzcd}$$

Q: How is d & d' related?

Def. (Hodge-Weil cycles)

Let (A,η) be an AV ~~with~~ of Weil-type.

We see that $H^1(A;\mathbb{Q}) \cong K^{\oplus n}$ so one gets a map

The natural map is in fact in opposite direction

$$\phi : K \cong \Lambda^n_K H^1(A;\mathbb{Q}) \hookrightarrow \Lambda^n_{\mathbb{Q}} H^1(A;\mathbb{Q}) = H^n(A;\mathbb{Q})$$

This map is not natural though.

The space of Hodge-Weil cycles is defined as

$$W_K := HW(A,\eta) := \operatorname{Im}\phi$$

<u>Def</u> (polarized AV of Weil-type)

(A,η,Λ) is of Weil-type, if the following diagram commutes:

$$\begin{tikzcd}
H^1(A;\mathbb{Q}) \times H^1(A;\mathbb{Q}) \arrow[r, "\Lambda"] \arrow[d, "\cdot\sqrt{-d}"'] \arrow[dr, dashed, "(\sqrt{-d})^*\Lambda"] & H^{2n}(A;\mathbb{Q}) \cong \mathbb{Q} \arrow[d, "\cdot d"] \\
H^1(A;\mathbb{Q}) \times H^1(A;\mathbb{Q}) \arrow[r, "\Lambda"] & H^{2n}(A;\mathbb{Q}) \cong \mathbb{Q}
\end{tikzcd}$$

not $(-d)$

Rmks. 1. be careful that $(\sqrt{-d})^*$ may have different meaning:

$(\sqrt{-d})^* = d \circ \eta(\sqrt{-d}) = \cancel{[\sqrt{-d}]}$ $[\sqrt{-d}]$. on $H^k(A;\mathbb{Q})$, not on Λ.

2. One can define a non-deg K-Hermitian form (+ with values in K):

$$H : H^1(A;\mathbb{Q}) \times H^1(A;\mathbb{Q}) \longrightarrow K$$

$$H(x,y) = \Lambda(x,(\sqrt{-d})^* y) + \sqrt{-d}\,\Lambda(x,y)$$

Check: $H(ax,y) = \bar{a} H(x,y)$.

General Strategy: When (A,η) is of Weil-type, try to show

① $B^p(A) = \langle D^p(A), W_K \rangle$

② $W_K \subset \mathrm{Im}\, cl^p$

Then

$$B^p(A) \supset \mathrm{Im}\, cl^p \supset \langle D^p(A), W_K \rangle \quad \Rightarrow HC \checkmark$$

① Thm. When $n=4$, we have basically 3 cases: (up to isogeny)

$\left[\begin{smallmatrix}2+2,\\2+1+1\end{smallmatrix}\right]$ (1) [RM08] $A = X_1 \times X_2$ of abelian surfaces, then HC ✓

(I don't know if ① ② are true in this case)

[3+1] (2) If $A = E \times X$, X simple, $B^2(A) \neq D^2(A)$, then

$\Downarrow$ [MZ99, Prop 3.8] + [vG04, Thm 6.6]

$\mathrm{End}^\circ(E) = K \subset \mathrm{End}^\circ(X)$, $K \in CM_2$,

$\Downarrow$ [MZ99, Thm 0.1(i)]

$B^2(A) = \langle D^2(A), W_K \rangle$

[4] (3) [MZ95, Thm 2.11]

If A is simple of Weil-type with $K \in CM_2$, then

$B^2(A) = \langle D^2(A), W_K \rangle$.

Thm [MZ 99. Thm 0.2]. When $n=5$, we have the following cases. (up to isogeny)

[4+1] (1) $A = E \times X$, X simple.

(1a) when $\mathrm{End}^\circ(E) = K \subset \mathrm{End}^\circ(X)$, $K \in CM_2$, and K acts on T_0X with mult $(1,3)$, then

$$B^\cdot(A) = D^\cdot(A).$$

(1b) otherwise,

$$B^\cdot(A) = B^\cdot(E) \times B^\cdot(X) = \mathrm{Im}\, cl^\cdot$$

[3+1+1] (2) $A = E^2 \times X$, X simple with $\mathrm{End}^\circ(E) = K \subset \mathrm{End}^\circ(X)$, $K \in CM_2$, then

$$B^\cdot(A) = \langle D^\cdot(A), \alpha^* W_K \rangle_{alg} \qquad \alpha : A \longrightarrow E_2 \times X$$

[3+1+1] (3) $A = E_1 \times E_2 \times X$, X simple with $\mathrm{End}^\circ(E_2) = K \subset \mathrm{End}^\circ(X)$, $K \in CM_2$, $E_1 \not\sim_{iso} E_2$, then

$$B^\cdot(A) = \langle D^\cdot(A), \alpha^* W_K \rangle_{alg} \qquad \alpha : A \longrightarrow E_2 \times X$$

(4) Otherwise,

$$B^\cdot(A) = D^\cdot(A).$$

② Thm [Markman 25, Thm 1.5.1]

For $n=6$, A: Weil-type with $K \in CM_2$, $\det H = -1$,

$$W_K \subset \mathrm{Im}\, cl^p$$

$\xrightarrow{\text{[Schoen 98, Prop 10]}}$

For $n=4$, A: Weil-type with $K \in CM_2$,

$$W_K \subset \mathrm{Im}\, cl^p.$$